Prevention
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Introduction
Title

بسم الله الرحمن الرحيم

Cases
Drugs
Pathophysiology
Risk Factors
Introduction

Drug-induced lung diseases



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More than 600 drugs are known to cause pulmonary toxicity. This number is expected continue to increase as new therapeutic agents and illicit drugs are developed and used.

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Drug-induced pulmonary toxicity is a diagnosis of exclusion. Other disease processes must be ruled out before establishing this diagnosis. Such diseases processes include respiratory infections; occupational, recreational, and environmental exposures; specific respiratory disorders; and systemic diseases.

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Adverse drug reactions can involve the pulmonary parenchyma, pleura, airways, pulmonary vascular system, mediastinum, and neuromuscular system that contribute to respiration. These reactions can manifest acutely, subacutely, or chronically.

TABLE 66-1 General Principles of Drug-Induced Lung Injury

Clinical presentation is nonspecific
Injury occurs with variable latency from drug initiation
Lung injury is often dose-independent
Pulmonary toxicity may be unrelated to the drug's pharmacologic properties
Acute, subacute, and chronic reactions may be caused by a drug
A variety of histopathologic patterns may be induced by a drug
Diagnosis of drug-induced injury is often made by exclusion
Resolution of injury may occur with drug discontinuation alone
Rechallenge with the suspected culprit drug is *not* recommended

The likelihood of developing adverse pulmonary effects from drugs remains largely unpredictable

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Some of the known risk factors for these selected drugs include the following:

The likelihood of developing adverse pulmonary effects from drugs remains largely unpredictable

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Some of the known risk factors for these selected drugs include the following:

- Advanced age
- Cumulative dose
- Simultaneous oxygen therapy
- Combination therapy

(Coadministration of cisplatin and bleomycin ,The combination of vinblastine and mitomycin increases the risk of asthma.)

- Radiation therapy
- Occupational factors (Asbestos & bleomycin)
- Underlying lung disease
- Genetic factors

Mechanisms of pulmonary injury

Pulmonary toxicity secondary to drugs may be due to a variety of mechanisms, such as the following:

- Oxidant injury.
- Pulmonary vascular damage.
- Deposition of phospholipids within cells.
- Immune system–mediated injury.
- Central nervous system (CNS) depression.
- Direct toxic effect.

Interstitial pneumonitis

Interstitial pneumonitis (ie, inflammation of the lu interstitium) is the most common manifestation of induced lung disease.

A wide array of drugs can cause interstitial pneumonitis of the agents implicated are azathioprine, bleomycin, N phenytoin, statins, amiodarone, and sulfasalazine.

NCPE

A variety of drugs can cause NCPE. This is a less common pattern of drug-induced involvement than pneumonitis and fibrosis.

Drugs can cause pulmonary edema by two mechanisms

- Capillary endothelial injury, leading to leakage of fl and protein into the lung interstitium.
- Certain drugs depress the central nervous system (CNS), resulting in neurogenic pulmonary edema.

NCPE

Clinically, acute NCPE manifests as an acute onset of dyspnea with tachypnea, tachycardia, hypoxemia, and diffuse crackles on physical examination. Fluffy infiltrates are present on the chest radiograph.

Drugs that cause NCPE include the following :

- Heroin
- Interleukin 2
- Methotrexate (MTX)
- Cocaine
- Tocolytic agents
- Hydrochlorothiazide
- Cyclophosphamide
- Iodinated radiographic contrast agents

Hypersensitivity reaction

Drug hypersensitivity results from interactions between a pharmacologic agent and the patient's immune system. These reactions are commonly associated with nitrofurantoin, MTX, beta-blockers, and procarbazine.

BOOP (Bronchiolitis obliterans-organizing pneumonia)

BOOP is a distinctive pattern of lung response to a few drugs. Histology reveals interstitial inflammation superimposed on the dominant background of alveolar and ductal fibrosis.

Drugs that can cause BOOP include the following:

- Acebutolol
- Amiodarone
- Amphotericin B
- Bleomycin
- Carbamazepine

Pulmonary vascular disease

A) Venous thromboembolism (DVT & PE)

The exact mechanism by which drugs increase the risk for thromboembolism is still unknown. But many theories may attribute to thrombus formation

- Direct endothelial damage and expose the underlying subendothelium thus leading to platelet adherence and subsequent thrombus formation. (e.g contrast media and cytotoxic drugs)

- Attenuate the secretion of pro- and anticoagulation mediators by the endothelial cells and may have prothrombotic effects on platelets by increasing adhesion and aggregation.(e.g COCs)
- Decreased blood flow by increasing blood viscosity. (e.g intravenous immunoglobulin administration)

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Other implicated drugs include

- Phenytoin
- Procainamide
- Retinoic acid

B) Pulmonary hypertension

Appetite suppressants (eg, amphetamines, fenfluramine, dexfenfluramine) are associated with an increased risk of pulmonary hypertension.

Prescription medications such as beta-blockers and mitomycin C have also been reported to cause pulmonary hypertension.

C)Pulmonary Vasculitis/ hypersensitivity pneumonitis

Drugs that cause pulmonary vasculitis, including nitrofurantoin, sulfonamides, penicillins, phenytoin, and propylthiouracil.

D) Diffuse alveolar hemorrhage

Patients usually present with hemoptysis, dyspnea, and hypoxemia. Diffuse alveolar hemorrhage is characterized by bilateral infiltrates in the context of anemia of recent onset and hypoxemia.

Drugs include

- Anticoagulants
- Cytosine arabinoside
- MTX
- Carbamazepine
- Cocaine,
- Hydralazine
- Mitomycin C
- Penicillamines.

Bronchospasm

Aspirin and other nonsteroidal anti-inflammatory drugs (NSAIDs) can induce bronchospasm (8-20% of adults with asthma). The acute asthmatic reaction occurs within 20 minutes to 3 hours after ingestion of the drug.

Bronchospasm has also been reported with the use of the following agents:

- Inhaled pentamidine/ acetylcysteine.
- Amphotericin B
- Amiodarone
- Angiotensin-converting enzyme (ACE) inhibitors (ACEIs)
- Beta-blockers
- ARVT for HIV.

Pleural involvement

Pleural effusions can develop in patients undergoing treatment with MTX, nitrofurantoin, amiodarone, procarbazine, carmustine, or cyclophosphamid.

Pleural effusions can also occur in drug-induced lupus; medications in this setting include hydralazine, procainamide, phenytoin, nitrofurantoin, and ACEIs.

Positive serum testing for antinuclear and histone antibodies aids in the diagnosis of this disorder.

Bilateral pleural thickening is a distinctive form of late cyclophosphamide toxicity.

Pneumothorax can complicate late stages of drug-induced pulmonary changes and has been reported in association with bleomycin, carmustine, and retinoic acid.

Mediastinal involvement

- Enlargement of hilar and mediastinal lymph nodes may be produced by phenytoin, bleomycin, and carbamazepine can induce.
- Pseudosarcoidosis syndrome can develop with interferon alfa and beta.

- Mediastinal lipomatosis

It is the accumulation of excess unencapsulated fat within the mediastinum. It may be seen in patients with Cushing disease or those treated with corticosteroid therapy.

The usual appearance on the chest radiograph is a smooth widening of the anterior and superior mediastinum without any deformity of the trachea. The fat pads in the costophrenic angles are also often enlarged. The diagnosis is made based on chest computed tomography (CT) scan findings.The treatment is cessation of steroid therapy.

ug	Pulmonary Syndrome	Treatment	Comments
omycin	Chronic pneumonitis; pulmonary fibrosis; rare fulminant variant with acute respiratory failure	Discontinue drug; corticosteroids	Risk factors: Cumulative dose >400 units; supplemental oxygen; therapeutic radiation; renal insufficiency; age >70 y; additional cytotoxic drugs
	Hypersensitivity-type lung disease	Discontinue drug; corticosteroids	Dyspnea, cough, skin rash, and peripheral eosinophilia
	Chest pain syndrome	Discontinue drug	Associated with drug infusion
tomycin C	Chronic pneumonitis; pulmonary fibrosis; rare fulminant variant after single dose	Discontinue drug; corticosteroids	Risk factors: Oxygen therapy; therapeutic radiation; additional cytotoxic drugs May relapse after withdrawal of corticosteroids
	Acute dyspnea and/or bronchospasm	Discontinue drug; bronchodilators; supportive care; consider corticosteroids	Risk factors: Vinca alkaloid use; may develop noncardiogenic pulmonary edema and/or chronic interstitial disease Can recur with rechallenge of vinca alkaloids
	Hemolytic uremic syndrome	Discontinue drug; supportive care; poor response to plasmapheresis or corticosteroids	Microangiopathic hemolytic anemia thrombocytopenia, renal insufficiency, noncardiogenic pulmonary edema; rare hemorrhage
tinomycin D	"Radiation recall" pneumonitis	Discontinue drug	Effect may be longstanding

	Pulmonary Syndrome	Treatment	Comments
amide	Chronic pneumonitis; pulmonary fibrosis	Discontinue drug; corticosteroids	Risk factors: High cumulative dose; additional cytotoxic drugs; supplemental oxygen; therapeutic radiation Toxicity may occur years after treatment is completed Pleural fibrosis may occur in late-stage disease
	Chronic pneumonitis; pulmonary fibrosis	Discontinue drug; consider corticosteroids	Risk factors: Cumulative dose >500 mg Often occurs after cessation of drug; prognosis poor
	Eosinophilic pneumonitis; organizing pneumonia; diffuse alveolar damage	Discontinue drug; corticosteroids	Risk factors: Pre-existing interstitial lung disease
	Chronic pneumonitis; pulmonary fibrosis	Discontinue drug; consider corticosteroids for patients with hypersensitivity features	Clinical pulmonary toxicity is rare with exception of high-dose melphalan (>200 mg/m^2)

	Pulmonary Syndrome	Treatment	Comments
otrexate	Chronic pneumonitis; pulmonary fibrosis	Corticosteroids; discontinue drug	Most common syndrome of methotrexate-induced lung toxicity; risk factors: older age, pre-existing lung disease, diabetes, previous use of disease-modifying drugs, hypoalbuminemia
	Hypersensitivity-type lung disease	Corticosteroids; discontinue drug	May resolve even if drug is continued, but can progress to fibrosis
	Acute chest pain syndrome	Discontinue drug	Often accompanied by pleural effusions
	Noncardiogenic pulmonary edema	Supportive care; discontinue drug	Associated with intrathecal administration
ine noside	Noncardiogenic pulmonary edema	Supportive care; discontinue drug	Onset of symptoms usually occurs within days of initiation of treatment; risk factor: cumulative dose
	Cryptogenic organizing pneumonia	Discontinue drug; corticosteroids	Risk factors: combined therapy with anthracyclines or interferon-α
rabine	Hypersensitivity-type lung disease; interstitial pneumonitis	Discontinue drug	Toxicity is uncommon; treatment associated with increased incidence of delayed opportunistic infections
itabine	Dyspnea	Occurs within hours of dose	Usually self-limited
	Interstitial lung disease; noncardiogenic pulmonary edema	Discontinue drug; corticosteroids; diuretics	Risk factors: combination treatment with taxanes or bleomycin

TABLE 65-5 Diagnosis of Methotrexate Pneumonitis

Diagnostic criteria:

- Acute onset of shortness of breath
- Fever (>38.0°C)
- Tachypnea (≥28 breaths/min) with nonproductive cough
- Radiographic evidence of interstitial or alveolar infiltrates
- WBC ≤15,000
- Negative blood or sputum cultures for pathogenic organisms (required)
- Pulmonary function tests demonstrating restrictive disease with low diffusion capacity
- Pa_{O_2} <55 mm Hg on room air (at presentation)
- Biopsy histopathology consistent with bronchiolitis or interstitial pneumonitis with giant cells and without evidence of pathogenic microorganisms

Presence of methotrexate pneumonitis:

- Definite: at least 6 of 9 criteria
- Probable: 5 of 9 criteria
- Possible: 4 of 9 criteria

Drug	Pulmonary Syndrome	Treatment	Comments
Carmustine (BCNU)	Early-onset interstitial lung disease	Discontinue drug; corti-costeroids	Risk factors: total dose, female sex, underlying lung disease, cigarette smoking, combina-tion treatment with other cytotoxic drugs
	Late-onset pulmonary fibrosis	May appear years after treatment	
Lomustine Semustine Chlorozotocin	Interstitial lung disease	Discontinue drug	By extrapolation, toxicities and risk factors probably parallel BCNU

Drug	Pulmonary Syndrome	Treatment	Comments
Gefitinib	Interstitial lung disease; pulmonary fibrosis	Discontinue drug	Usually occurs within first 3 mo of treatment; risk factors: older age, cigarette smoking, pre-existing lung disease, Asian ethnicity
Erlotinib	Interstitial lung disease	Discontinue drug	Risk factors: pre-existing lung disease
Imatinib	Pleural effusions	Diuretics	
	Interstitial lung disease	Discontinue drug; corticosteroids	Risk factors: pre-existing interstitial lung disease
Dasatinib	Pleural effusions	Diuretics	
	Pulmonary arterial hypertension	Discontinue drug	
Bevacizumab	Pulmonary hemorrhage	Supportive care; discontinue drug	Risk factors: cavitary tumor, squamous cell histology
Cetuximab	Bronchospasm, stridor	Discontinue drug; supportive care; corticosteroids	May occur during infusion; risk factors: asthma, atopy, history of allergic reactions
Rituximab	Interstitial pneumonitis; crypto-genic organizing pneumonia	Discontinue drug; corticosteroids	
	Bronchospasm, stridor, angioedema	Discontinue drug; supportive care; corticosteroids	May occur during infusion
Trastuzumab	Bronchospasm, hemodynamic instability	Discontinue drug	May occur during infusion
	Interstitial lung disease	Discontinue drug; corticosteroids	

Case (1)

Male patient 55 years old is known to have IHD maintained on ACE inhibitors, statins and aspirin. The patient presented to our outpatient clinic complaining of persistent dry cough over the past 6 weeks.

Q: Mention your possible diagnosis and your management.

Answer 1:

The most possible Dx is Aspirin-induced asthma or dry cough mediated by ACE inhibitors.

Cessation of ACE inhibitors and its replacement with ARBs is a better choice.

Giving antihistamine, leukotriene receptor blocker with a bronchodilator can help to minimize symptoms.



Case 2 :

35 years old female patient admitted to Respiratory intensive care unit (RICU) due to massive pulmonary embolism. The patient received Streptokinase followed by I.V heparin as anticoagulant. 2 days later, the patient developped haemoptysis, anaemia, hypoxia and bilateral pulmonary infiltrates.

Q: Give the possible diagnosis.

Answer 2 :

The most possible diagnosis is diffuse alveolar haemorrage caused by anticoagulants given.

Case 3 :

An 80-year-old woman complains of dry cough and progressive SOB over the past 4 weeks. Her past medical Hx includes rheumatoid arthritis 5 yrs ago maintained on prednisolone 10 mg/day and MTX 15 mg/wk and she had started anti-TNFa therapy 3 months ago. The patient is tachypnic,hypoxic (SaO2 at rest 81% on room air)

Q: what is your differential diagnosis.

Answer 3 :

- Methotrexate:

✓ Sub acute interstitial pneumonitis

- Corticosteroids :

Increase the risk for pnemonia

- Anti-TNFa:

Increase risk for pneumonia



Methods

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Aim of the work

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